

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths *	Prac **
5	(a)	(i)		Make sure maggots do not come into contact with NaOH/ stop the maggots from drowning			1	1		1
		(ii)		Any two from - Gives time for the maggots to acclimatise to temp/conditions/ owtte(1) - Gas may expand/contract(1) - Time for {NaOH to equilibrate/equipment to equilibrate/ a {constant/ 25 °C} temperature} (1)		2		2		2
		(iii)		- Carry out same experiment under same conditions with same {mass/volume } of {glass beads/ dead disinfected maggots}(1) - Shows it is the respiration of maggots causing the (volume) change (1)			2	2		
	(b)	(i)		Any two from: EXPT1. - Oxygen used by the maggots for <u>aerobic</u> respiration; (1) <u>Released</u> CO₂ is absorbed by NaOH (1) - Volume/pressure decreases in tube. (1) EXPT2. Volume of O ₂ used equal to volume of CO ₂ evolved (1)			3	3		3
		(ii)		- increases {reliability / reproducibility} (of mean) (1) NOT accurate/ repeatability - increase confidence / {exclude/ identify} anomalous results / allows use of {statistical test/ named statistical test} (1)		2		2		2

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5	(b)	(iii)		27.6 mm ³ g ⁻¹ minute ⁻¹ = 3 marks 27.58 = 2 marks (not to 1 dp) 110.3 = 2 marks (Using diameter instead of radius) 110.32 = 1 mark (Using diameter and not to 1dp) 137.9 = 2 marks (Using 5 minutes) 13.8 = 2 marks (Using 0.5g) 13.79 = 1marks (Using 0.5g and not 1 dp) Cross sectional area of lumen. $\pi r^2 = 1.13\text{mm}^2$ Volume used = $1.13 \times 61 = 68.93\text{mm}^3$ Divide by 5 to give per minute = 13.8mm^3 per minute. x 2 to give per g = $27.6 \text{ mm}^3 \text{ g}^{-1} \text{ minute}^{-1}$		3		3	3
	(c)	(i)		102/145 = 0.7/ 0.70/ 0.703		1		1	1
		(ii)		Lack of water in environment (1) Produce {a <u>lot of metabolic</u> water/ a lot of water from respiration} (1)			2	2	
		(iii)		Glycogen does not need a large oxygen supply/ ORA (1) And does not need to remove a large amount of CO ₂ /water/ ORA (1)			2	2	
				Question 5 total	0	8	10	18	4 10